How to Install Python (Mac and Windows) - v2

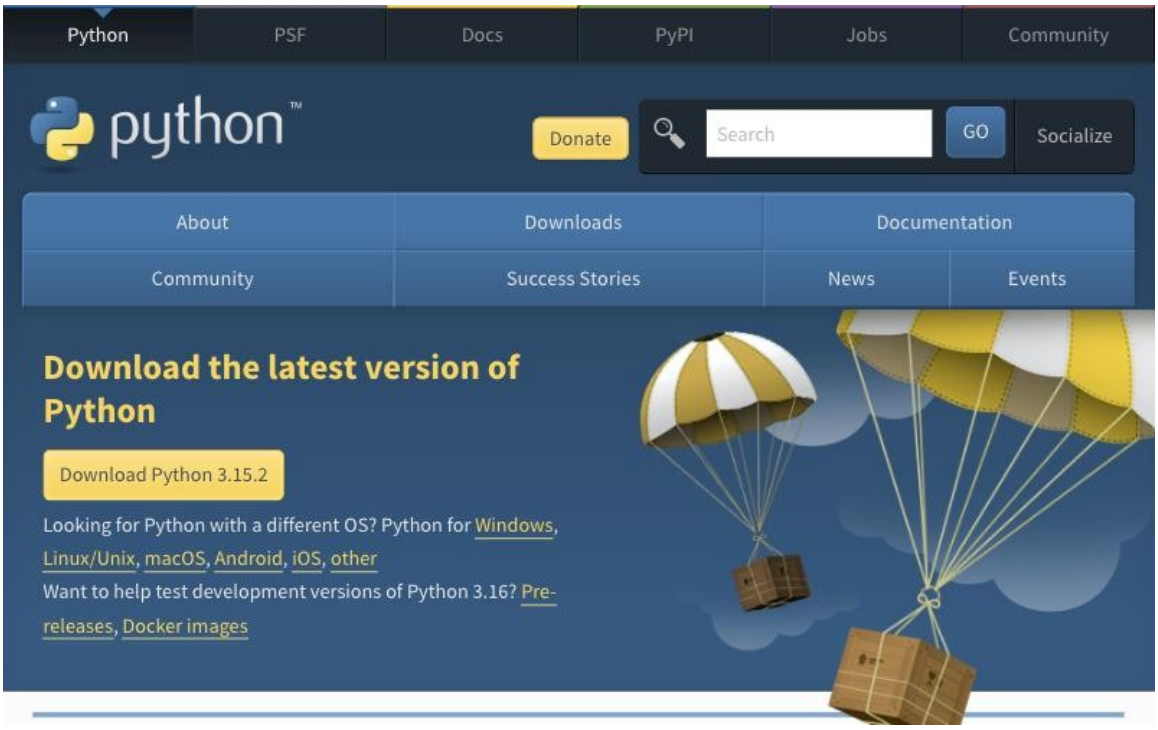
Version 2. Screenshots were captured on 2026-11-10 from the official python.org website, which showed Python 3.15.2 as the latest release (released 2026-10-25) and 3.14.10 as a stable release. Versions will change over time. This guide has no dependency on any workshop files.

1. Before you start

- Check whether Python is already installed. In a terminal (macOS Terminal; Windows Command Prompt or PowerShell) run the lines below. A version number means Python is installed.
- If your instructor names a specific Python version, install that one. Newest releases can be too new for some scientific packages; if unsure, a 3.14.x release (listed as stable on python.org) is a conservative choice.
- Operating system requirements differ between releases. Read the requirement shown on the python.org page for the release you choose (the Windows and macOS download pages and each release page list it). Python 3.15 supports Windows 10 and newer.

```
python3 --version      # macOS
python --version       # Windows (or: py --version)
```

Note for Mac users: macOS may include its own /usr/bin/python3 (installed with the Xcode Command Line Tools). It is often old and used by Apple tools. Do not delete or modify it; install the python.org version alongside it.



python.org/downloads: the yellow button downloads the latest Python for your operating system. Links below it lead to the Windows and macOS pages.

2. Install on macOS

1. Go to python.org/downloads and click the "Download Python <version>" button (or open python.org/downloads/macos and choose a release). The file is an installer package (.pkg) that runs on both Apple silicon and Intel Macs.

2. Double-click the downloaded .pkg file.

3. Click Continue through Introduction and Read Me, then click Continue and Agree on the licence.

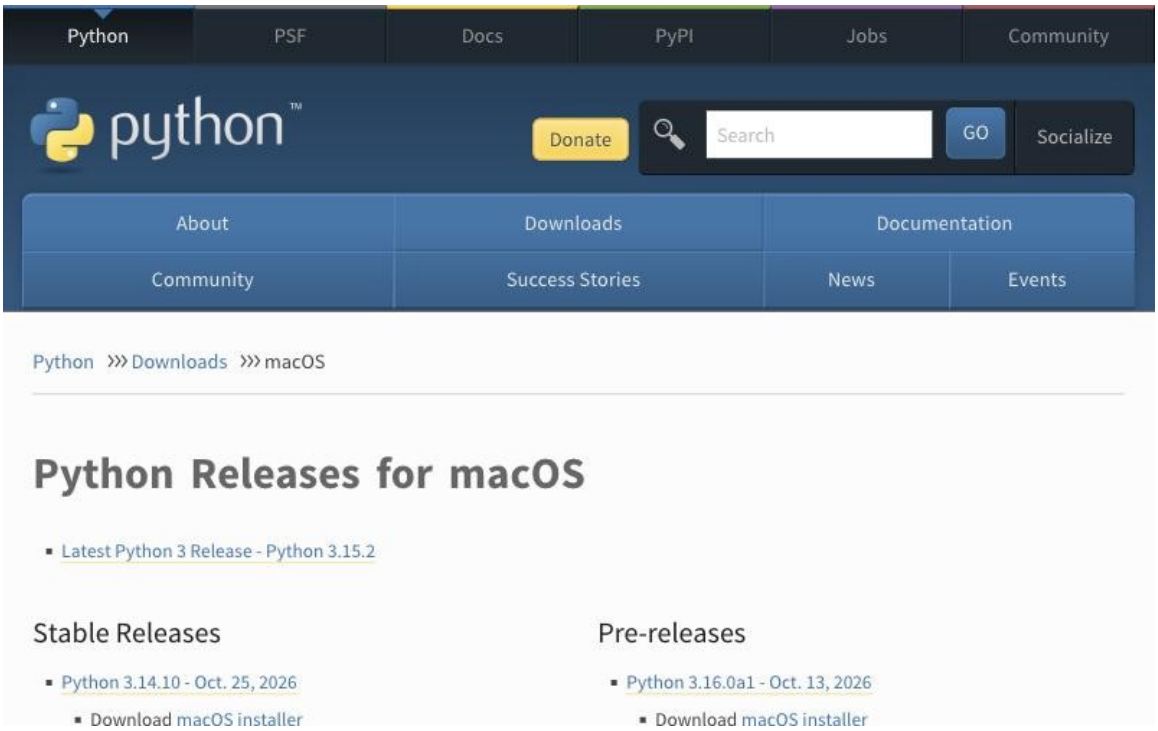
4. Click Install (the default options are fine). Enter your Mac password when asked; installing requires an administrator account.

5. When the installation finishes, a window titled Python <version> opens. Double-click Install Certificates.command. A Terminal window runs and should end with "update complete". Close it. This step lets Python connect securely to websites.

6. Open a new Terminal and check the install (replace 3.15 with the version you installed):

```
python3 --version
python3.15 --version
```

The installer also adds IDLE (a simple editor) and Python Launcher to /Applications/Python <version>/. If python3 still shows an older version, use the version-specific command, such as python3.15, or open a fresh Terminal window.



python.org/downloads/macos lists the latest release and earlier stable releases, each with a "macOS installer" link.

Alternative: if you already use Homebrew (brew.sh), you can run "brew install python". Package managers are community maintained and may lag behind the newest release.

3. Install on Windows

python.org now recommends the Python install manager for Windows. It gives you the commands python, py and pymanager and makes installing or updating Python versions easy.

1. Go to python.org/downloads and click the download button, or open python.org/downloads/windows and choose "Latest Python install manager". (You can also install it from the Microsoft Store: search for "Python install manager".)

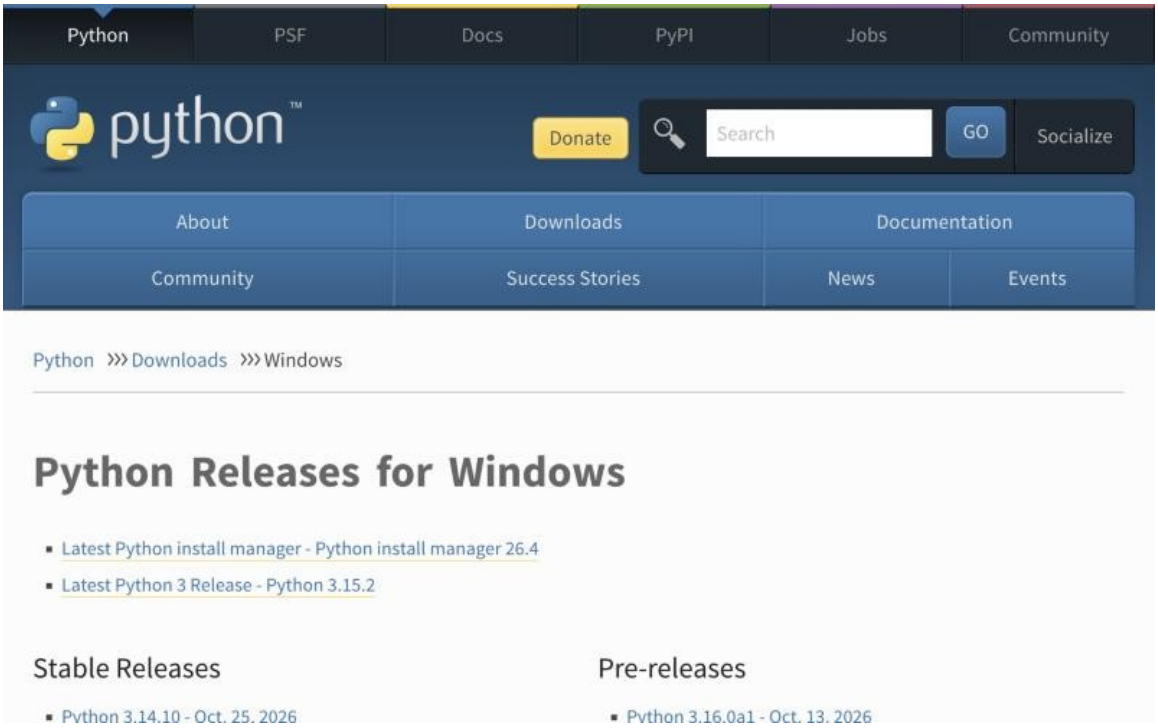
2. Double-click the downloaded file (an .msix package) and click Install. If Windows says it cannot install it, use the Microsoft Store version.

3. Open Command Prompt or PowerShell. This is the terminal; its prompt looks like C:\Users\you> or PS C:\Users\you>. Type py --version and press Enter. The first time, the install manager may offer to install the latest Python runtime and to add its folder to your PATH. Accept the install; adding to PATH is optional if you use the py command.

4. Python has its own prompt, which looks like >>> . If you ever see it, you are inside Python, not in the terminal. Type exit() and press Enter to return to the terminal.

5. Back at the terminal prompt, verify the install:

```
python --version
py --version
py list
```



python.org/downloads/windows: the "Latest Python install manager" link and the latest Python 3 release.

Standalone installer (alternative): python.org also offers a classic installer (.exe) for a specific release. If you use it, tick "Add python.exe to PATH" on the first screen, then choose Install Now. The python.org documentation notes that this .exe installer is deprecated since Python 3.14, so prefer the install manager if you can.

4. Create a virtual environment (recommended)

A virtual environment is a separate toolbox for one project, so packages from different projects do not clash. Run these in your project folder. Use the activation line for your own shell.

```
# macOS (Terminal)
python3 -m venv .venv
source .venv/bin/activate

REM Windows - Command Prompt
python -m venv .venv
.venv\Scripts\activate.bat
```

```
# Windows - PowerShell
python -m venv .venv
.\.venv\Scripts\Activate.ps1

# Windows - Git Bash
python -m venv .venv
source .venv/Scripts/activate
```

When it is active, your prompt shows (.venv). Leave it with the command: deactivate. In VS Code choose it with Command Palette > Python: Select Interpreter.

PowerShell may refuse to run the activation script ("running scripts is disabled"). On a computer managed by your school or employer, do not try to bypass the policy. Either use Command Prompt (activate.bat) or skip activation and call the environment's Python directly, which works the same way:

```
.\.venv\Scripts\python.exe --version
.\.venv\Scripts\python.exe -m pip install <package>
```

If neither works, ask your IT support or teaching team.

5. Troubleshooting

- Windows: typing python opens the Microsoft Store or says not found: open Start, search "Manage app execution aliases", and make sure the Python aliases are turned on (try switching them off and on).
- Windows: py says "can not open file": uninstall the old "Python launcher" from Installed apps so the install manager is used.
- Windows: the next commands "do nothing" or show errors: check whether the prompt starts with >>> ; if so type exit() first.
- macOS: "command not found: python": use python3 (macOS does not provide a command named python).
- macOS: SSL or certificate errors when downloading packages: run Install Certificates.command from the /Applications/Python <version> folder.
- "pip" not found: use python -m pip install <package> (macOS: python3 -m pip).
- Several Pythons installed: on Windows py list shows them all; py -V:3.14 chooses one.